# A Portable AI-Assisted Screening System for Fine Motor Developmental Delay in Preschool Children Using PDMS-2 Tasks

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## ABSTRACT

Early identification of fine motor developmental delay enables timely intervention, yet standardized assessments such as the Peabody Developmental Motor Scales, Second Edition (PDMS-2) require trained therapists, who are scarce in rural areas. This study presents a portable screening system that automates the administration and scoring of PDMS-2 fine motor items for children aged 3–6 years. The system integrates an assemblable testing booth with top-view and side-view cameras, a touchscreen that delivers task instructions, and standardized test materials. Seventeen items in five task categories (block building, shape drawing, paper cutting, paper folding, and pellet placement) are scored by a multi-stage pipeline that combines YOLOv8n object detection, Segment Anything (SAM, ViT-B) segmentation, convolutional classification, and rule-based geometric measurement, while an Arduino-based sensor records pellet placement in real time. Results are stored in a MySQL database for longitudinal tracking. A clinical trial evaluating the system's screening accuracy against occupational therapist ratings is currently underway at Chang Gung Memorial Hospital, Tucheng Branch. The system can be set up by non-specialist staff in about five minutes, suggesting its potential as a low-barrier screening tool for community health centers in resource-limited areas.

**Keywords:** PDMS-2, fine motor delay, computer vision, early screening

## 1. INTRODUCTION

Developmental delay in early childhood arises from interacting biological and environmental risk factors; early intervention during the preschool years improves children's functional outcomes and family well-being [1]. Delays are usually identified through routine primary-care surveillance or parent/teacher observation [2], and structured occupational-therapy-led activities have been shown to improve preschoolers' fine motor performance [3].

Access to assessment, however, is uneven: rural children are less likely to receive timely early intervention [4], and in rural Taiwan, long travel distances and a shortage of pediatric therapists hinder evaluation [5]. Taiwan introduced six additional government-funded developmental screenings for children under seven in July 2024 [6], increasing demand for screening tools usable outside hospitals.

The PDMS-2 [7] is a widely used motor assessment with good internal consistency and test–retest reliability [8], but its administration requires trained examiners, limiting use in community settings without therapists. Computer vision now makes it feasible to automatically analyze the products of children's motor tasks—block structures, drawings, and cut paper.

This study presents a portable screening system automating the administration and scoring of 17 PDMS-2 fine motor items. Its contributions are (1) an assemblable testing booth deployable by non-specialist staff, (2) a multi-stage AI scoring pipeline following the PDMS-2 scoring criteria, and (3) a cloud record system supporting longitudinal follow-up.

## 2. METHODS

### 2.1 System Architecture

The system comprises a testing booth, standardized test materials, an analysis server, and a web-based feedback interface (Fig. 1). The booth is an arch-shaped enclosure assembled from lightweight panels and was co-designed with the university's Department of Industrial Design. LED light strips inside the booth provide uniform illumination for image analysis, a dual-lens camera captures top-view and side-view images, and a 14-inch touchscreen presents the task instructions. A test mat fixes the positions of the child and the materials, and the blocks, drawing templates, and paper sheets replicate the dimensions specified in the PDMS-2 manual. All equipment fits in a single carrying bag and is designed to be set up in about five minutes.

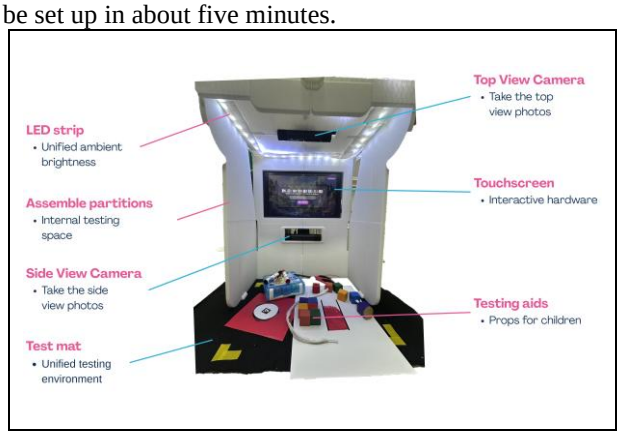



Figure 1. The portable testing booth

LED light strips, top-view camera, touchscreen, side-view camera, test mat, and test materials.

The software is a Flask web application with a MySQL database hosted on a laboratory server and accessed remotely through a Tailscale virtual private network. Access is divided into four roles: child (test taker), parent (views their own child's results), examiner (administers tests), and clinician/administrator (manages accounts, queries records, and exports reports).

### 2.2 Test Items

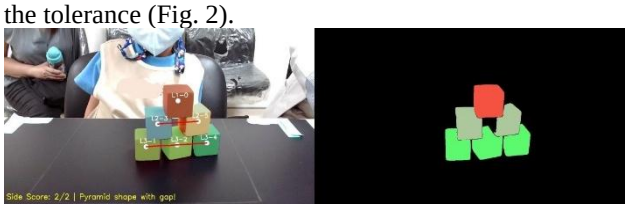
Seventeen PDMS-2 fine motor items suitable for children aged 3–6 years (36–71 months) were grouped into five task categories: block building (C1, four items), shape drawing (C2, six items), paper cutting (C3, four items),

paper folding (C4, two items), and pellet placement (C5, one item). Each task is introduced by a narrated animation and a demonstration video, so the examiner only needs to follow on-screen prompts and requires no occupational therapy background. Scores are not shown to the child.

### 2.3 AI Scoring Pipeline

Instead of a single end-to-end model, each item is scored by a pipeline that proceeds through detection, segmentation, classification, and measurement, and the final measurement is mapped to the PDMS-2 three-level score (2 = criterion met, 1 = partially met, 0 = not met) [7]. Object detection uses YOLOv8n (Ultralytics) [9], and fine segmentation uses the Segment Anything Model (SAM, ViT-B checkpoint) [10].

**Block building (C1).** YOLOv8n locates each block as a bounding box in the top or side view, and SAM refines each box into an instance mask. Masks are clustered into layers by their vertical centroid coordinates, and structural rules are checked. For the pyramid item, the structure must contain six blocks in three layers whose horizontal centers are aligned within a tolerance; a structure whose inter-block gaps are below a threshold scores 2, one with larger gaps scores 1, and any other configuration scores 0. In the side-view output, each block mask is colour-coded by its detected layer and labeled with its layer/index (e.g., "L1-0"), and a red line marks any inter-block gap that exceeds the tolerance (Fig. 2).

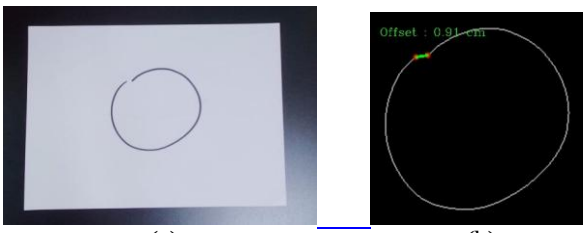



(a) (b)

Figure 2. Example AI Scoring Output for the Block-Building Pyramid Item (C1)

(a) the side-view photo, annotated with per-layer colour overlays, block/layer index labels, and a red gap-indicator line; (b) the isolated per-layer instance masks on a black background; the side score is 2/2 (pyramid shape with gap).

**Shape drawing (C2).** For the line-drawing, coloring, and dot-connection items, the A4 worksheet is located with a U-Net model (ResNet-34 encoder) and rectified by perspective correction; for the circle, square, and cross items, the worksheet is instead located by Canny edge detection and contour fitting before the same perspective correction is applied. Both paths yield a pixel-to-centimeter ratio. YOLOv8n then crops each drawn figure, a convolutional classifier (a plain CNN, not a pretrained backbone) implemented in TensorFlow distinguishes circles, ovals, quadrilaterals, and crosses, and skeletonization with endpoint analysis evaluates line continuity, closure, and deviation. For the circle item, the drawn line is skeletonized onto a black background: its endpoints are marked in red, the gap between them in green, and the measured offset (or "Perfect !" if the loop is fully closed) is printed in green (Fig. 3, 4).

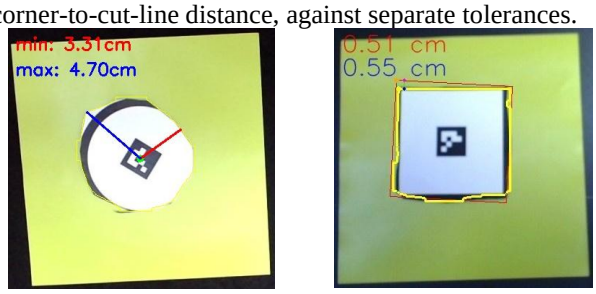



(a) (b)

Figures 3 and 4. Example AI Scoring Output for the Shape-Drawing Circle Item (C2)

(a) the original drawing, showing a small unclosed gap near the top of the circle; (b) the skeletonized trace with the endpoint gap highlighted, measuring an offset of 0.91 cm (score 2).

**Paper cutting (C3).** A YOLOv8n segmentation model extracts the cut-paper mask directly (no separate SAM step), an ArUco fiducial marker [11] provides metric calibration, and the minimum and maximum distances from the cut contour to the reference center are compared with the PDMS-2 tolerance for the circle-cutting item. For the square-cutting item, the same ArUco calibration instead measures the maximum edge deviation between the cut contour and the target square, and the maximum corner-to-cut-line distance, against separate tolerances.




(a) (b)

Figure 5. Example AI Scoring Output for the Paper-Cutting Item (C3)

(a) the circle-cutting item, calibrated with the ArUco marker (green dot), showing the minimum and maximum distances from the reference center to the cut contour (red 3.31 cm, blue 4.70 cm); (b) the square-cutting item, showing the maximum edge deviation (red 0.51 cm) and the maximum corner-to-cut-line distance (blue 0.55 cm).

**Paper folding (C4).** For the single-fold item, a YOLOv8n segmentation model masks the folded sheet directly and polygon fitting (approxPolyDP) measures the fold; for the double-fold item, colored-edge detection and largest-quadrilateral fitting instead locate the folded sheet. Geometric measurements of the fold line and the resulting shape determine the score.

**Pellet placement (C5).** An Arduino-based sensing container counts the pellets dropped in and reports the count and the remaining time to the server through a serial link; the score is determined by the number of pellets placed within the time limit.

Because clinical training images are scarce, the training sets were expanded with conventional augmentation (rotation and translation).

### 2.4 Records

Per-item scores, raw images, and annotated result images are stored in MySQL and served through signed URLs, allowing clinicians to review each child's history remotely.

### 2.5 Validation Protocol

Children aged 3–6 years are being recruited at Chang Gung Memorial Hospital, Tucheng Branch. Each child will complete all tasks in the booth while a licensed occupational therapist independently scores every item according to the PDMS-2 manual, and agreement between AI and therapist scores will be assessed with percentage agreement and Cohen's weighted kappa (κ) once data collection is complete. The study was approved by the Institutional Review Board of Chang Gung Medical Foundation (approval number 202500117B0), and written informed consent will be obtained from all parents.

## 3. RESULTS

A complete prototype integrating the booth, AI models, database, and web interfaces was implemented. A field deployment was conducted at a community care center on May 9, 2026, in which children operated the system in a real-world setting. Module-level performance on the project's own test images is summarized in Table 1: the segmentation-based pipelines (block building, paper cutting, paper folding) reached a mean IoU of 0.93–0.98, and the shape-drawing CNN classifier reached 0.90–0.97 accuracy, depending on the item.

Table 1 summarizes the scoring method, model accuracy, and the mean analysis time per task category (design target: ≤ 15 s per task). Paper-cutting and paper-folding items (C3, C4) ran well within the target, whereas block-building items (C1) exceeded it on every item, and shape-drawing items (C2) exceeded it in three of six items. These are single-run, CPU-only measurements from an offline batch test on the project's own test images (n = 1 per item); C5 (sensor-based) was not timed this way, as its score is produced instantaneously from the pellet count. Model accuracy in Table 1 is reported as mean IoU for the segmentation-based pipelines and as classification accuracy for the shape-drawing classifier; these two metrics are not interchangeable and are therefore not aggregated into a single figure.

Table 1. Scoring methods, model accuracy, and mean analysis time for each task category.

| Cat. | Task | Scoring method | Time (s) | Model Accuracy |
|---|---|---|---|---|
| 1 | C1-t1 String blocks | YOLOv8n + SAM + layer rules | 18.6 | IoU 0.93/0.97 (side/top); shared n=21/22 across C1 |
| | C1-t2 Pyramid | | 24.6 | |
| | C1-t3 Stair | | 22.2 | |
| | C1-t4 Build wall | | 28.0 | |
| 2 | C2-t1 Circle | Canny + YOLOv8n + CNN + skeleton | 25.1 | Acc 0.97 (n=32) |
| | C2-t2 Square | | 20.1 | Acc 0.95 (n=78) |
| | C2-t3 Cross | | 15.5 | Acc 0.90 (n=42) |
| | C2-t4 Line | U-Net (ResNet-34) + skeleton | 13.7 | IoU 0.99 (n=24); shared U-Net paper-segmentation step across C2-t4/5/6 |
| | C2-t5 Color | U-Net (ResNet-34) + colour range | 6.0 | |
| | C2-t6 Connect dots | U-Net (ResNet-34) + skeleton | 10.8 | |
| 3 | C3-t1 Cut circle | YOLOv8n-seg + ArUco + center distance | 2.4 | IoU 0.98 (n=19); shared across C3-t1/t2 |
| | C3-t2 Cut square | YOLOv8n-seg + ArUco + edge/corner distance | 6.6 | |
| | C3-t3 Cut paper | YOLOv8n-seg + ArUco + contour | 2.7 | IoU 0.96 (n=10); shared across C3-t3/t4 |
| | C3-t4 Cut line | YOLOv8n-seg + area ratio | 5.5 | |
| 4 | C4-t1 One fold | YOLOv8n-seg + polygon fit | 2.4 | IoU 0.98 (n=24); shared across C4-t1/t2 |
| | C4-t2 Two fold | Edge detection + quadrilateral fit | 2.7 | |
| 5 | C5-t1 Pellet placement | Arduino count + timer | N/A | N/A (sensor-based) |

## 4. DISCUSSION AND CONCLUSION

The proposed system turns a therapist-administered assessment into a guided procedure that non-specialist staff can conduct in community settings. Decomposing each item into detection, segmentation, and rule-based measurement keeps the scoring logic close to the PDMS-2 criteria and makes errors traceable, which is important for clinical acceptance. This study has limitations: clinical validation against occupational therapist ratings is still ongoing, only a subset of PDMS-2 fine motor items is covered, part of the training data is synthetic, and the block-building pipeline (C1, YOLO + SAM) exceeds the 15 s per-task target on CPU hardware. Future work will optimize the C1 pipeline, complete the ongoing clinical trial, expand the item set and age range, conduct multi-site validation, and route below-cutoff results to occupational

therapists through a LINE-based referral service. On the project's own test images, the YOLOv8n+SAM block-detection pipeline (C1) reached a mean IoU of 0.93 in the side view (n = 21) and 0.97 in the top view (n = 22); the YOLOv8n-seg cut-paper detector (C3) reached a mean IoU of 0.98 for the circle/square items (n = 19) and 0.96 for the other cutting items (n = 10); and the fold detector (C4) reached a mean IoU of 0.98 (n = 24). The shape-drawing CNN classifier (C2) reached 0.97, 0.95, and 0.90 accuracy for the circle, square, and cross items, respectively (n = 32, 78, 42). These are module-level segmentation/classification metrics from an internal test set, not the clinical agreement with occupational therapist ratings, which the ongoing clinical trial will assess separately.

In conclusion, the portable AI-assisted system provides a low-barrier approach to early screening of fine motor delay and has the potential to help community health centers in resource-limited areas identify children who need further evaluation and timely referral. A clinical trial to validate its accuracy against occupational therapist assessments is currently underway.

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